Decoding Parameter & Sampling Analysis

Effect of Temperature, Top-K and Top-P on the Output Consistency, Hallucination and Structure of Local LLMs (Ollama)

Models: Llama 3.2 3B (llama3.2:3b) and Qwen 2.5 3B (qwen2.5:3b)

Experiment: 2,160 generations, run on Google Colab (T4 GPU) through Ollama

Date of report: 21 October 2026

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Code, data and results: GitHub repository *llm-sampling-analysis* (link:[github.com/Habiba578/llm-sampling-analysis])

Abstract

This project measures how three decoding settings, Temperature, Top-K and Top-P, change the output of two small locally run language models. We sent 30 prompts (10 factual, 10 repeated-consistency, 10 structured-JSON) to Llama 3.2 3B and Qwen 2.5 3B through Ollama under 12 settings combinations, three times each with different seeds, giving 2,160 generations with no failed or empty output. Outputs were scored for run-to-run consistency, factual accuracy, hallucination and JSON validity.

The clearest finding is that consistency across seeds drops when Temperature rises from 0.2 to 0.7 or 1.0 (exact agreement of three runs: Llama 53% to 32%, Qwen 65% to 42%). We found no clear effect of Top-K (20 vs 40) or Top-P (0.8 vs 0.95) in this design. Factual accuracy was 100% everywhere, so it could not separate the settings. Hallucination depended mostly on the model, not on the settings: Qwen produced hallucination-type answers to the three probe prompts far more often than Llama. JSON was valid in all Llama outputs and in 94.2% of Qwen outputs, with most Qwen failures on one prompt. The results come from a small experiment (30 prompts, 3 runs, two 3B models) and should be read with the limitations in Section 8.

1. Introduction

A language model produces text one token at a time. At each step it assigns a probability to every possible next token, and a decoding (sampling) rule chooses one. Three common settings control this choice:

- **Temperature** rescales the probabilities. Low values make the most likely token even more likely; high values flatten the distribution, so less likely tokens are chosen more often.
- **Top-K** keeps only the K most likely tokens and discards the rest before sampling.
- **Top-P** (nucleus sampling) keeps the smallest set of tokens whose probabilities add up to P, and discards the rest.

These settings are often changed by trial and error. Our aim is to measure, with a repeatable local experiment, what they do to the reliability of a small model's answers.

2. Problem Statement and Objectives

Problem: how do Temperature, Top-K and Top-P influence (a) output determinism, (b) hallucination and (c) structural consistency of locally run LLMs?

- Build a Python benchmark that runs the same prompts under controlled settings and records every output.
- Measure run-to-run consistency, factual accuracy, hallucination-related errors and JSON validity.
- Compare the two models and find which settings give more consistent and reliable output.

3. Methodology

3.1 Models, tools and libraries

Item	Detail
Models	Llama 3.2 3B (llama3.2:3b) and Qwen 2.5 3B (qwen2.5:3b), downloaded with Ollama

Runtime	Ollama on Google Colab, T4 GPU
Python libraries	ollama (send prompts), pandas (data), tqdm (progress bar), matplotlib (graphs)
Request	ollama.chat with one user message and no system prompt; one attempt per generation

3.2 Dataset (30 prompts)

Category	IDs	Purpose
Factual	F01-F10	F01-F07: single-answer questions scored for accuracy. F08-F10: hallucination probes (F08 and F09 ask about an invented novel and an invented paper; F10 asks who first walked on Mars, a question with a false premise)
Repeated	R01-R10	Prompts used to examine consistency of answers across runs; R07 and R08 also have a correct answer
Structured	S01-S10	Prompts that require a JSON reply with given keys and value types

Each prompt has a reference answer and scoring instructions (designed in the experiment plan, docs/experiment_plan.docx). The titles and authors in F08 and F09 were invented for this study. A web search on 18 October 2026 found no exact match for either title or author, but a limited search cannot prove that no such work exists.

3.3 Parameter grid (12 configurations)

Temperature (0.2, 0.7, 1.0) x Top-K (20, 40) x Top-P (0.8, 0.95) gives 12 configurations, C01 to C12 (full table in Appendix A). Fixed for every request: maximum new tokens 256, context length 4096. Each prompt was run three times per model and configuration with seeds 42, 43 and 44. Total: 2 models x 12 configurations x 30 prompts x 3 runs = **2,160 generations**. All completed with no error and no empty response.

Because the seed differs between the three runs, the experiment measures **consistency across seeds**, not strict determinism under an identical seed. Whether the seed option gives identical output for identical settings was not tested.

3.4 Metrics

Metric	Definition
Exact agreement	For one (model, configuration, prompt), the three responses are equal after trimming spaces and ignoring case (F02 is case-sensitive because the answer is the symbol Au). Reported as the share of such groups in which all three agree.
Pairwise agreement	Share of the 3 response pairs that are equal (0, 1 or 3 of 3).
Word overlap	Average Jaccard similarity of the word sets of the 3 response pairs (0 to 1). This similarity method was chosen by the evaluation part of the project.
Factual accuracy	Share of responses to F01-F07, R07 and R08 whose text contains the reference answer.
Hallucination	F08-F10 responses labelled *correct* (admits it cannot verify the item, or rejects the false premise), *hallucination* (invented content presented as fact) or *mixed* (says it is not sure but still gives invented details). The main rate counts hallucination and mixed; the strict rate counts clear hallucination only.
JSON validity	Strict: the whole response parses as JSON. Lenient: also accepted if only a markdown code fence surrounds the JSON. Schema correct: exact keys and value types. Values correct: values match the reference.

Time	Wall-clock time per call. The first call of each model includes model loading and is excluded from medians.

The evaluation code is evaluation/evaluate.py. Accuracy and hallucination labels are decided by keyword rules; the way they were checked is described in Section 8.

4. Expected Results

Before the experiment we expected that: (1) low Temperature and small Top-K or Top-P give more consistent outputs; (2) higher Temperature gives more hallucination and more invalid JSON; (3) the two models behave somewhat differently.

5. Actual Results

All numbers below come from results/raw_outputs.csv (2,160 rows) and are saved as tables in evaluation/outputs/tables.

5.1 Run-to-run consistency

Model	Temperature	Exact agreement	Pairwise agreement	Word overlap
Llama 3.2 3B	0.2	53.3%	60.0%	0.839
Llama 3.2 3B	0.7	35.0%	42.2%	0.708
Llama 3.2 3B	1.0	31.7%	39.4%	0.708
Qwen 2.5 3B	0.2	65.0%	68.6%	0.881
Qwen 2.5 3B	0.7	43.3%	50.0%	0.739
Qwen 2.5 3B	1.0	41.7%	48.9%	0.706

Each row uses 120 groups (30 prompts x 4 configurations). Consistency is highest at Temperature 0.2 and falls at 0.7. The step from 0.7 to 1.0 is small (3 and 2 percentage points). With 120 groups the sampling uncertainty of one rate is about 4 to 5 percentage points, so the difference between 0.7 and 1.0 is not reliable, while the drop from 0.2 to 0.7 (18 and 22 points) is clearly larger. Even at 0.2, the three runs agree exactly in only 53% (Llama) and 65% (Qwen) of groups, because the seed changes between runs and many answers are free-form sentences.

Model	Top-K 20	Top-K 40	Top-P 0.8	Top-P 0.95
Llama 3.2 3B (exact agreement)	40.0%	40.0%	41.1%	38.9 %
Qwen 2.5 3B (exact agreement)	50.6%	49.4%	50.6%	49.4 %

Top-K and Top-P changed exact agreement by at most about 2 points, which is within the uncertainty. We therefore cannot claim an effect of either setting in this design. Only two values of each were tested, and both Top-K values are large; a wider range (for example Top-K 1 or Top-P 0.3) would be needed to test them properly.

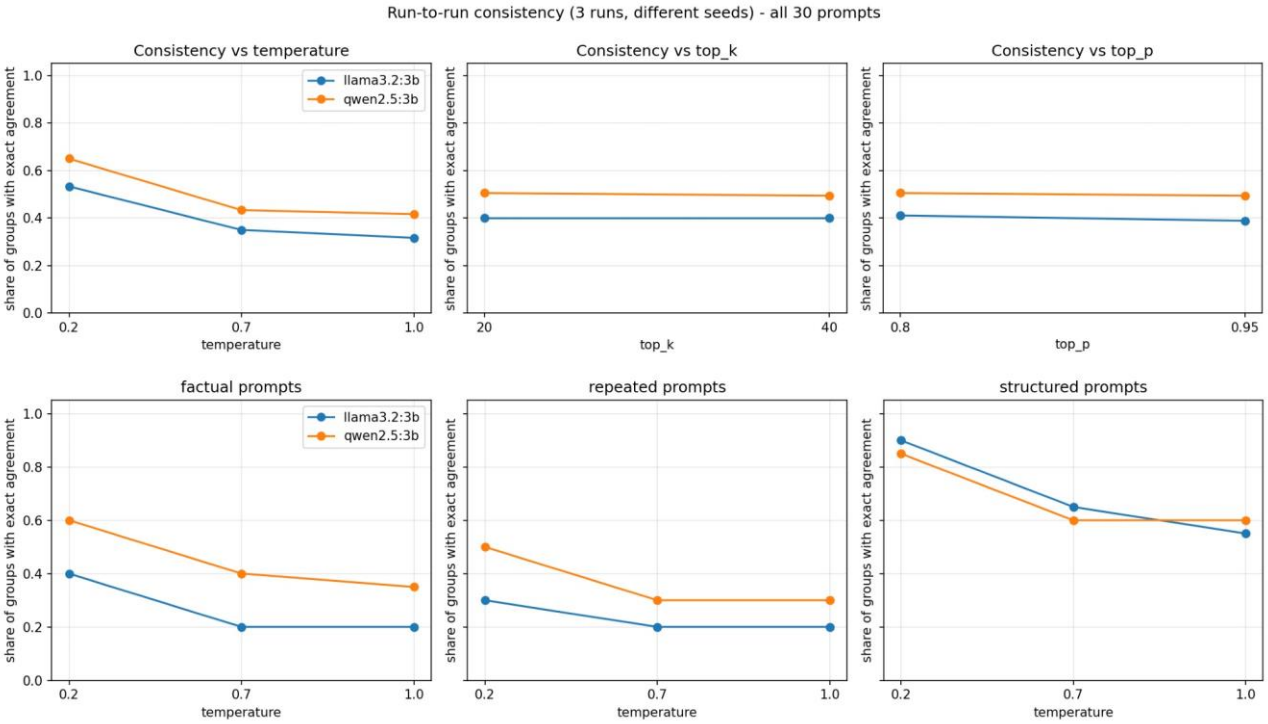


Figure 1. Share of groups in which all three runs agree exactly, by Temperature, Top-K and Top-P.

Model	Factual	Repeated	Structured
Llama 3.2 3B (exact agreement)	26.7%	23.3%	70.0 %
Qwen 2.5 3B (exact agreement)	45.0%	36.7%	68.3 %

Structured (JSON) answers are far more consistent than free-text answers because the format leaves little room for different wording.

Figure 2. Exact agreement by prompt category and Temperature.

5.2 Factual accuracy

Both models answered 324 of 324 accuracy-scored responses correctly (100%), at every Temperature. The prompts are easy facts (for example the capital of Australia or the square root of 144), so accuracy shows a ceiling effect and cannot separate the settings. This does not show that settings never affect accuracy; harder questions would be needed.

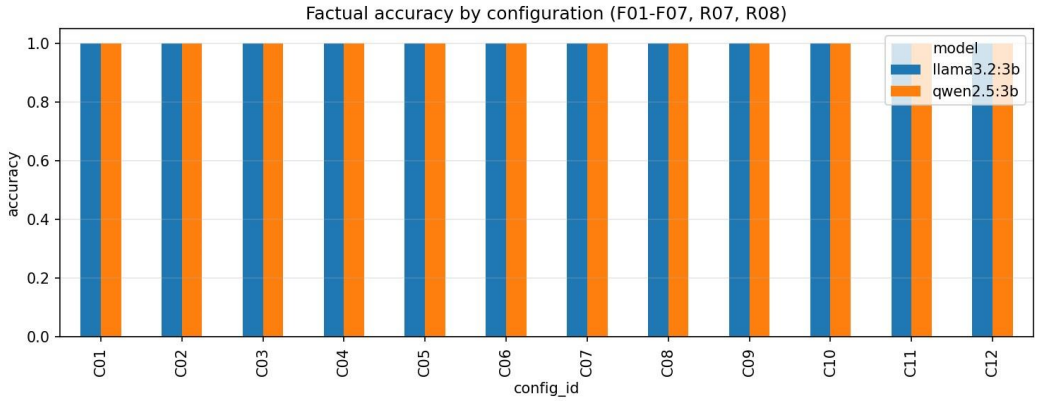


Figure 3. Factual accuracy by configuration (all 100%).

5.3 Hallucination

Model	Probe	Outputs	Correct	Mixed	Halluc.	Main rate	Strict rate
Llama 3.2 3B	F08 and F09 (invented items)	72	68	4	0	5.6%	0 %
Llama 3.2 3B	F10 (false premise)	36	36	0	0	0%	0 %
Qwen 2.5 3B	F08 and F09 (invented items)	72	41	2	29	43.1%	40.3 %
Qwen 2.5 3B	F10 (false premise)	36	0	31	5	100%	13.9 %

Main rate = hallucination + mixed; strict rate = clear hallucination only. The difference between the two rates depends on how *mixed* answers are judged. In F10, Qwen usually wrote that no human has walked on Mars yet but also named a NASA rover as "the first person to walk on Mars". We count this as mixed (a false answer is still given), which is why Qwen's main rate on F10 is 100% while its strict rate is 13.9%.

Both rates are reported so the reader can see the effect of this judgement.

On F08 (an invented novel) both models almost always said they could not find it. On F09 (an invented paper), Llama said it could not find it in 32 of 36 runs and in the other 4 added invented details after saying so; Qwen described the paper as if it existed in 29 of 36 runs.

Model	Temperature	Outputs	Main rate	Strict rate
Llama 3.2 3B	0.2	36	0%	0 %
Llama 3.2 3B	0.7	36	5.6%	0 %
Llama 3.2 3B	1.0	36	5.6%	0 %
Qwen 2.5 3B	0.2	36	66.7%	44.4 %
Qwen 2.5 3B	0.7	36	58.3%	22.2 %
Qwen 2.5 3B	1.0	36	61.1%	27.8 %

For Qwen the hallucination rate does **not** rise with Temperature (it is highest at 0.2). Llama's few mixed answers occurred only at 0.7 and 1.0, but these are 2 answers each. The size of the difference between the models (main rate 3.7% for Llama vs 62.0% for Qwen over all 108 probe responses each) is far larger than any effect of Temperature in our data. Only three probe prompts were used, so this result is about these prompts, not about hallucination in general.

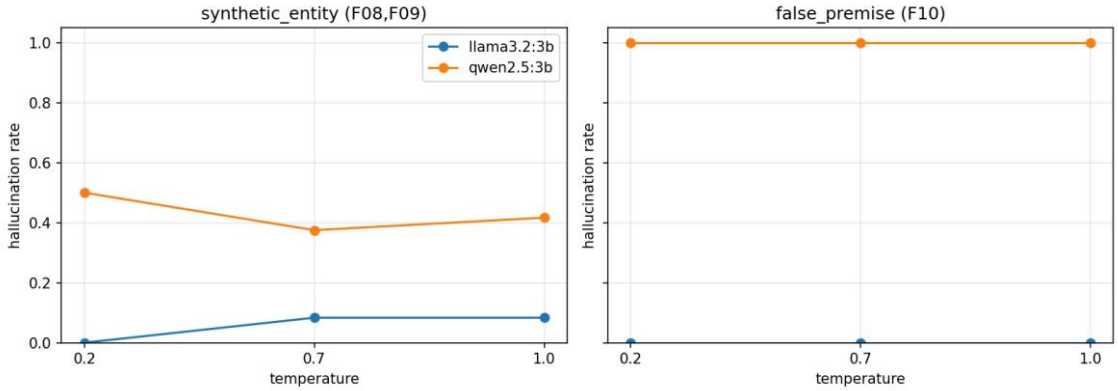


Figure 4. Hallucination rate (main rate) by Temperature and probe type.

5.4 Structured output (JSON)

Model	Strict valid	Lenient valid	Schema correct (among valid)	Values correct
Llama 3.2 3B	100%	100%	90.0%	88.9 %
Qwen 2.5 3B	94.2%	94.2%	85.0%	75.6 %

Lenient validity equals strict validity: neither model wrapped its JSON in a markdown code fence. Each model has 360 structured responses. The failures are concentrated in a few prompts (Figure 6):

- **S05 (Python release year):** both models returned the year as a string (Qwen "1991", Llama "1991-02-20") instead of a JSON integer, so the schema check failed in every run. The prompt did not say that the value must be an integer, so this partly reflects the prompt design.
- **S07 (array of three objects):** Qwen produced invalid JSON in 58.3% of runs, for example ["day": "Monday", ...] without braces. Llama was always correct on this prompt.
- **S06 (colour of a lemon):** Qwen gave a green or part-green colour in 44% of runs, which does not match the reference "Yellow".
- **S04 (position of Jupiter):** Llama gave a wrong position (2 instead of 5) in 4 of 36 runs.

Model	Temp 0.2	Temp 0.7	Temp 1.0
Llama 3.2 3B (strict validity)	100%	100%	100 %
Qwen 2.5 3B (strict validity)	90.8%	95.0%	96.7 %

Qwen's validity went slightly *up* with Temperature, the opposite of our expectation. Most Qwen failures come from one prompt (S07) and the changes are only a few responses, so we do not interpret this as a real effect of Temperature.

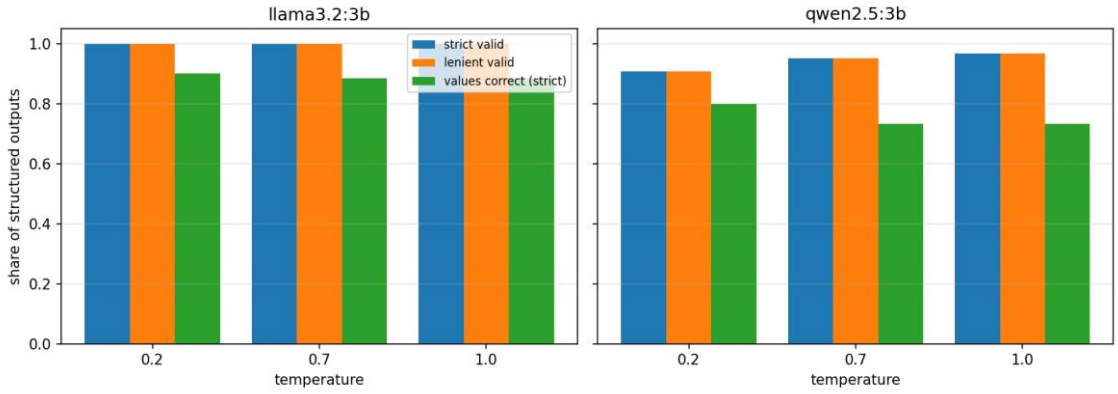


Figure 5. JSON validity and value correctness by model and Temperature.

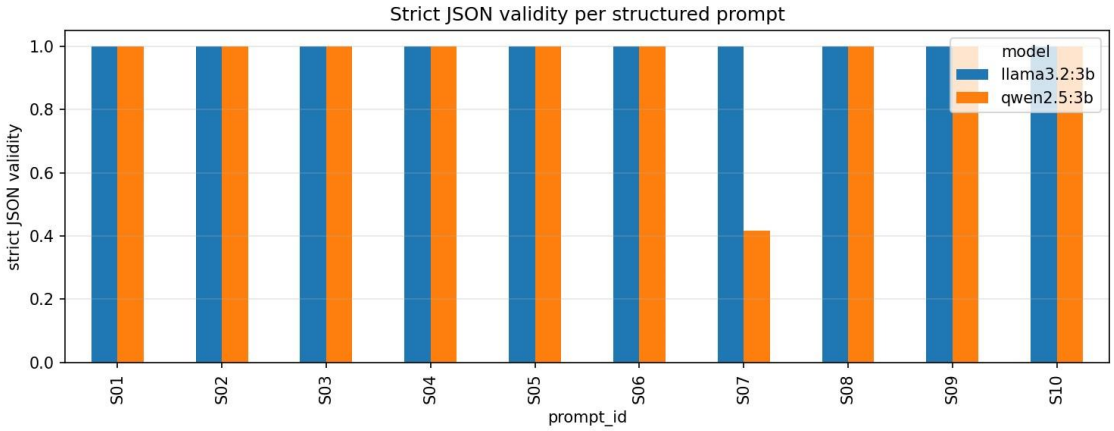


Figure 6. Strict JSON validity per structured prompt.

5.5 Time

After removing the first call of each model (model loading: about 88 seconds for Llama and 51 seconds for Qwen), the median time per call was 0.34 s for Llama and 0.32 s for Qwen, about 80 generated tokens per second for both on the T4 GPU. Median time rose slightly with Temperature (Qwen 0.30 s at 0.2 to 0.33 s at 1.0), which may be due to slightly longer outputs; we did not test this. The prompts are short, so these times only show that both models are fast on this hardware.

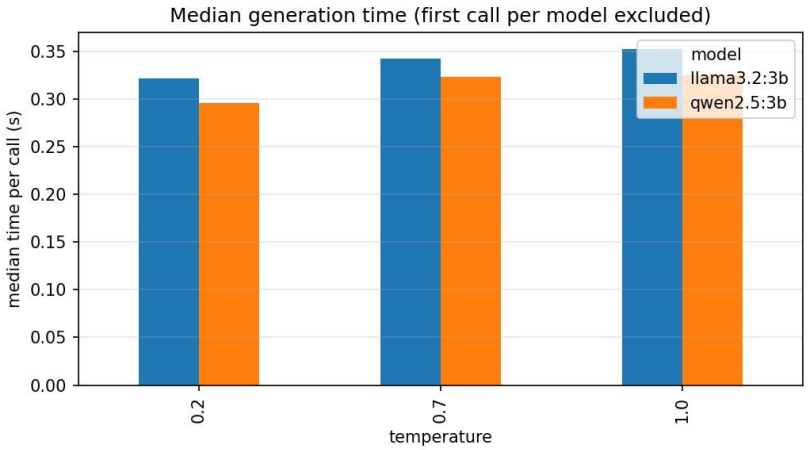


Figure 7. Median generation time by Temperature.

6. Discussion: Expected vs Actual

Expectation	What we found
Lower Temperature gives more consistent output	Supported. Exact agreement fell from 53% to 32% (Llama) and 65% to 42% (Qwen) between Temperature 0.2 and 1.0, mostly between 0.2 and 0.7.
Small Top-K and Top-P give more consistent output	Not shown. Differences were within about 2 points (Top-K 20 vs 40, Top-P 0.8 vs 0.95); the tested range was too narrow to conclude.
Higher Temperature gives more hallucination	Not shown. The hallucination rate did not rise with Temperature. The model (Llama vs Qwen) mattered far more.
Higher Temperature gives more invalid JSON	Not shown. Llama was always valid; Qwen's failures were mainly one prompt and did not rise with Temperature.
The models behave differently	Supported. Qwen was somewhat more consistent in wording, but hallucinated much more on the probe prompts and produced less correct JSON.

In this experiment, if consistency across runs matters, a low Temperature such as 0.2 is the safest of the three tested values; for the two 3B models and these prompts, the choice of model affected hallucination and JSON quality much more than Top-K, Top-P or Temperature did.

7. Conclusion

Using 2,160 generations from two local 3B models, we found that Temperature reduces run-to-run consistency (clearly from 0.2 to 0.7), that Top-K and Top-P (at the two values each tested) had no visible effect, and that accuracy on easy facts was unaffected. Hallucination and JSON quality differed mainly between models and prompts, not between settings. The study gives a repeatable benchmark and scripts, but larger prompt sets, more runs and a wider range of settings are needed before general rules can be stated.

8. Limitations

- **Small sample:** 30 prompts, three runs per prompt, two small (3B) models, one machine. Differences of a few percentage points are within chance.
- **Consistency, not determinism:** the seed differed between runs. Whether identical seeds give identical output was not tested.
- **Limited settings range:** three Temperatures and two values each of Top-K and Top-P. Top-K 1 or very low Top-P were not tested.
- **Accuracy ceiling:** the factual questions were too easy to separate the settings.
- **Few hallucination probes:** three prompts (F08-F10); F08 and F09 use invented items that a limited web search did not find, but absence cannot be proven. The mixed-answer judgement changes the rates (Section 5.3).
- **Automatic labels:** accuracy and hallucination are decided by keyword rules. The 75 answers flagged by the rules and a random sample of 20 factual and structured answers were read and checked, and two rule errors were found and corrected. This review was done with all member.
- **Truncation:** 42 answers reached the 256-token limit and were cut off (mainly F09 and R04). No JSON answer was cut off.
- **Prompt design:** S05 did not state that the value must be an integer, which caused schema failures in both models.
- **Environment:** the Ollama version and the models' default settings (ollama show) were not recorded.

9. Reproducibility and Repository

Everything is in the GitHub repository. To repeat the experiment: install Ollama; run *ollama pull llama3.2:3b* and *ollama pull qwen2.5:3b*; run *pip install -r requirements.txt*; run *python run_experiments.py* (reads data/, writes results/); then run *python evaluation/evaluate.py*.

Path	Content
data/	prompts_dataset.csv / .json (30 prompts), parameter_grid.csv (12 configurations)
docs/	experiment_plan.docx and the dataset design README
run_experiments.py	Generation script (resumes if stopped; saves timing fields)

results/	raw_outputs.csv and raw_outputs.json (2,160 generations, unedited)
evaluation/	evaluate.py, MANUAL_REVIEW_LOG.md, outputs/ (scored data, 20 tables, 7 figures)
report.pdf	This report

Appendix A. Parameter grid

Config	Temperature	Top-K	Top-P	Max tokens	Context	Seeds (run 1, 2, 3)
C01	0.2	20	0.8	256	4096	42 , 43, 44
C02	0.2	20	0.95	256	4096	42 , 43, 44
C03	0.2	40	0.8	256	4096	42 , 43, 44
C04	0.2	40	0.95	256	4096	42 , 43, 44
C05	0.7	20	0.8	256	4096	42 , 43, 44
C06	0.7	20	0.95	256	4096	42 , 43, 44
C07	0.7	40	0.8	256	4096	42 , 43, 44
C08	0.7	40	0.95	256	4096	42 , 43, 44
C09	1.0	20	0.8	256	4096	42 , 43, 44
C10	1.0	20	0.95	256	4096	42 , 43, 44
C11	1.0	40	0.8	256	4096	42 , 43, 44
C12	1.0	40	0.95	256	4096	42 , 43, 44

Appendix B. Scoring rules in brief

- Accuracy: the response must contain the reference answer (F01 Canberra, F02 Au, F03 1945, F04 12, F05 Shakespeare, F06 Jupiter, F07 6, R07 102, R08 cold).
- F08 and F09: correct if the response says it cannot find or verify the item; mixed if it says so but then states invented details; hallucination if it describes the item as if real.
- F10: correct if it rejects the premise without naming a person; mixed if it rejects the premise but still names a "first person" (for example a rover); hallucination otherwise.
- JSON: the exact required keys, correct types (integer, boolean, string, number) and reference values; S10 only needs answer = 14 and a confidence number between 0 and 1.